Algebra 1
Unit 5
Lesson 5 Practice

Name Answer Key
Date
Period

Jayne and Tolu are each raising money for charity for their club at school. They decide to make it a friendly competition. Jayne's club is hosting a walk-a-thon. She collects a total of $\$35$ for every mile she walks. Tolu's club is holding a carwash. He earns $\$25$ for charity for every car he washes, and one customer gave him a $\$100$ donation and did not even get a car wash!

1. Define a variable to represent the unknown.
   x= number of completed tasks

2. Write an expression for the amount of money Jayne raised for her club.
   $35x$

3. Write an expression for the amount of money Tolu raised for his club.
   $25x + 100$

4. When the events are over, they realized that they both completed the same number of tasks and earned the same amount of money. Set the two expressions equal to each other to create an equation.
   $35x = 25x + 100$
   $10x = 100$

5. Solve the equation and interpret your answer.
   $10x = 100$
   $x = 10$
   They completed ten tasks each.

6. How much money Jayne and Tolu each raise?
   \$350 each

7. If Jayne earned $\$37$ per mile instead, would it have been possible for the two friends to earn the same amount of money for the same number of tasks?
   $37x = 25x + 100$
   $12x = 100$
   $x = \frac{25}{3}$ or $8\frac{1}{3}$

   No, because you can't have a fractional task.

Tran is hungry for lunch, but he is on a budget. Two local Mexican restaurants have affordable lunch specials. Leo Tapatio charges $\$2$ per taco and $\$5$ for a plate of rice and beans and a drink. Taco Loco charges $\$1$ per small street-taco and has an all you can eat salad and salsa bar for $\$7.50$.

8. Define a variable to represent the unknown.
   x= number of tacos

Tran wants to spend the same amount of money at each restaurant for the same number of tacos, so created and solved an equation. His work is shown below.

$$2x + 5 = x + 7.5$$
$$2x + 5 - x = x + 7.5 - x$$
$$x + 5 = 7.5$$
$$x + 5 - 5 = 7.5 - 5$$
$$x = 2.5$$

9. Interpret Tran's answer.
   2.5 tacos would cost the same at either restaurant.

10. Explain whether Tran's solution to the equation is viable in this context.
    No, because you can't buy half a taco.